Mechanistic Interpretability of GPT-like Models on Summarization Tasks

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Project Goal & Motivation

Large language models such as GPT-2 generate strong summaries but operate as black boxes. We expose their inner workings—tracking how attention heads, neuron circuits and low-rank (LoRA) updates reshape information flow during fine-tuning—so practitioners can link model decisions to summarisation quality and trust.

Dataset & Implementation details

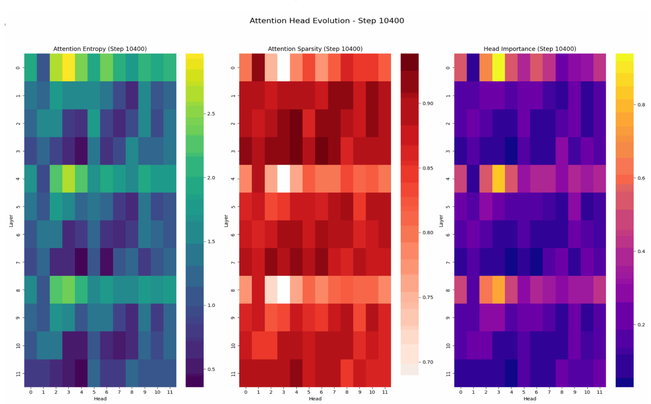
- CNN/DailyMail v3.0.0 (287 k train / 13 k val / 11 k test) fine-tunes GPT-2 (124 M) for summarisation.
- The same hyper-params are reapplied with a rank-8 LoRA adapter, letting us contrast full and parameter-efficient tuning head-to-head.

Component	Description / Setting
Dataset	CNN/DailyMail v3.0.0 287,113 train / 13,368 val / 11,490 test
Model	GPT-2 (124 M parameters)
Learning Rate	3×10^{-5}
Optimizer	AdamW (weight decay 0.01)
Batch Size	32
Epochs	10 (early-stopping patience 6)
GPU Resources	NVIDIA A100 (40 GB) / A16 (16 GB)
Token Limit	1024 input tokens, 128 output

Proposed Method

Interpretability Analysis

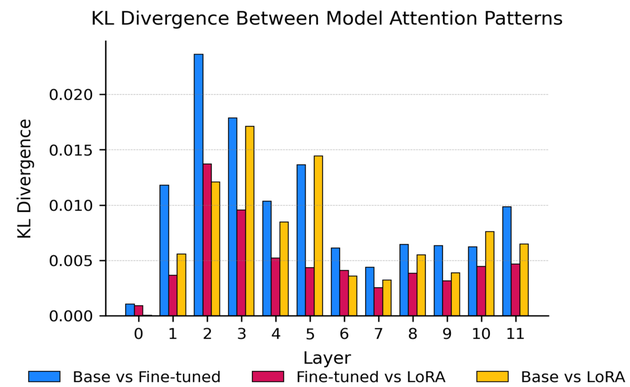
We quantify head-level reconfiguration by computing attention entropy $H(A) = -\sum_i a_{i\backslash \log} a_i$ and a sparsity index derived from L_1/L_2 ratio across all query–key pairs. Jointly minimising entropy and maximising sparsity after fine-tuning reveals emergence of a low-cardinality head-set that carries the dominant mutual-information mass, isolating the functional "summarisation circuit" within layers 5–8.



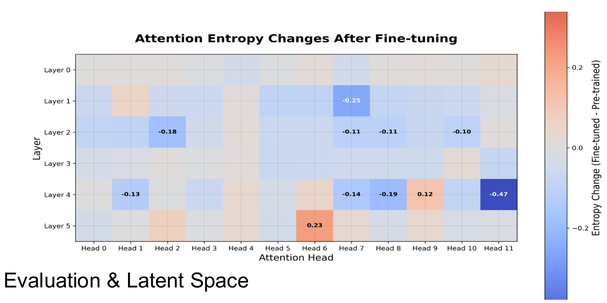
Head-Level Attention Reconfiguration: Entropy, Sparsity, and Importance Maps

Experiment Results

Quantitative Interpretability Findings : KL Divergence



Layer-wise KL divergence Layer-wise Kullback–Leibler divergence reveals the exact strata where adaptation rewires attention: sharp peaks in layers 2–4 for Base → Fine-tuned, milder co-peaks for Fine-tuned → LoRA, and near-zero elsewhere—showing LoRA reproduces early-layer re-weighting while leaving upper layers unchanged. $D_{\{KL\}(P \mid Q)} = \sum_{\{x \in X\}} p(X) \log(\frac{p(X)}{q(X)})$

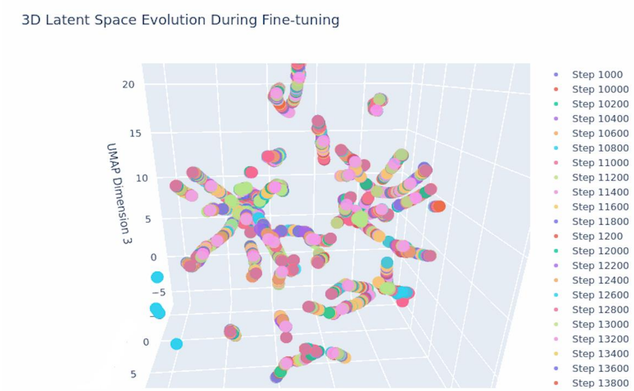


Evaluation & Latent Space

ROUGE evaluates summary fidelity by computing recall-oriented n-gram overlap. so ROUGE-1 measures unigram coverage, ROUGE-2 bigram coverage, and ROUGE-L uses longest-common-subsequence recall to factor in global ordering. Higher values therefore indicate that the generated summary captures more of the reference's lexical and structural content with minimal penalties for paraphrase.

Model	ROUGE 1	ROUGE 2	ROUGE L
Base (zero-shot)	0.137	0.057	0.102
Fine-tuned	0.186	0.089	0.123
LoRA-tuned	0.148	0.050	0.105

ROUGE 1, ROUGE 2, ROUGE L



UMAP Trajectory of Token Embeddings Across Fine-Tuning Steps

Progressive colour shifts show embeddings converging into compact manifolds, marking where semantic structure stabilises during training. Tracking this contraction pinpoints checkpoints for deeper circuit probes and explains performance jumps in later interpretability metrics.

Conclusion & Summary

Layer-wise probes reveal that full fine-tuning concentrates attention entropy and spikes KL-divergence in layers 2–6, yielding the highest ROUGE gains; LoRA reproduces the same reconfiguration with <1 % parameter change, making it a lightweight yet interpretable surrogate. These results demonstrate that on-the-fly interpretability can pinpoint precisely which strata to target for efficient task adaptation. Next, we will port the pipeline to LLaMA-2/3, investigating how rotary position embeddings (RoPE) reshape these circuits and whether low-rank adapters still capture the RoPE-induced geometry. We will further validate causal pathways via activation-patching fidelity scores and release an interactive visualization toolkit to let practitioners explore these findings in real time.

References

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